

PREDICTIVE MAINTENANCE AGENT FOR MANUFACTURING

Complete Project Definition and Implementation Blueprint

Sector: Manufacturing (generic)

Dataset: Predictive Maintenance Dataset (AI4I 2020)

Core parameters: Air Temperature, Process Temperature, Rotational Speed, Torque, Tool Wear

Machine data -> Condition analysis -> Failure prediction -> Failure mode -> Explanation -> Maintenance recommendation

This document is self-contained and is written specifically for the new predictive-maintenance project. It does not depend on or describe any previous project.

1. Problem Overview

Manufacturing machines can experience unexpected failures that interrupt production. When a machine fails, production output can fall, delivery schedules can be affected, maintenance costs can increase, and available machine capacity can be lost.

Traditional monitoring can show current readings or alert when a value crosses a threshold. However, monitoring alone does not answer an important question: **Is the machine likely to fail soon, and what is the likely failure mode?**

The proposed system addresses this gap by using historical machine operating data to learn failure patterns and provide an early warning before a failure causes significant downtime.

Problem statement

Manufacturing needs a system that can use machine operating data to detect abnormal behavior, predict failure risk, identify the likely failure mode, explain the prediction and recommend a suitable maintenance action.

2. How Big Is the Problem?

Unexpected equipment downtime is a major operational problem because a stopped machine can interrupt production and create secondary effects such as delayed orders, idle labour, increased maintenance effort, additional operating cost and reduced productivity.

The problem becomes more difficult when organizations depend on manual inspection or reactive maintenance. In a reactive approach, maintenance is performed after a failure or after a serious abnormal condition is observed. Predictive maintenance aims to identify risk earlier so that maintenance can be planned.

Impact area	Effect of unexpected machine failure
Production	Loss of output and reduced machine availability.
Cost	Lost production and potentially higher emergency maintenance costs.
Delivery	Production interruptions can affect order completion and delivery commitments.
Maintenance	Reactive failures can require urgent inspection, repair or replacement.
Productivity	Machine capacity is under-used while equipment is stopped.
Planning	Unexpected failures make production and maintenance planning harder.

For the final presentation, organization-specific downtime cost figures should only be included if supported by a reliable source. The AI4I dataset itself is a predictive-maintenance dataset, not a study of the financial cost of manufacturing downtime.

3. Who Is Facing the Problem?

The project is intentionally kept at the **generic manufacturing sector** level. We are not restricting the solution to automotive, textile, food, electronics or any other single manufacturing sub-sector.

Primary users

- **Factory managers:** need an overall view of machine health and failure risk.
- **Maintenance teams:** need early warnings and information about what to inspect.
- **Production supervisors:** need to know which machines may interrupt production.
- **Machine operators:** need simple and understandable warnings.
- **Plant management:** needs historical failure trends and maintenance priorities.

Why generic manufacturing is appropriate

The problem statement is about predictive maintenance for manufacturing machines in general. The AI4I 2020 dataset is based on a milling process and can therefore be used as a representative demonstration dataset without restricting the entire project to one industry sub-sector.

4. What We Are Building

We are building a **Predictive Maintenance Agent for Manufacturing**. The agent analyzes machine operating parameters, calculates condition indicators, detects abnormal patterns, predicts the probability of failure, identifies the likely failure mode and recommends maintenance.

Main objectives

- Analyze the core machine parameters.
- Calculate meaningful derived indicators.
- Detect abnormal operating conditions.
- Predict whether a machine is likely to fail.
- Identify the most likely failure mode.
- Explain which parameters contributed to the prediction.
- Recommend an appropriate maintenance action.

**Core flow: Machine data -> Calculations -> Conditions -> Failure prediction -> Failure mode -> Explanation
-> Maintenance recommendation**

5. Dataset: AI4I 2020

The selected dataset is the **Predictive Maintenance Dataset (AI4I 2020)**, commonly available through Kaggle and the UCI Machine Learning Repository. It contains 10,000 observations and 14 columns and represents a synthetic milling-process environment.

The dataset includes machine operating parameters and failure labels. Our core predictive inputs are the five numerical machine parameters listed below.

6. Parameters We Are Considering

Parameter	Unit	Role in the system
Air Temperature	K	Thermal/ambient operating condition.
Process Temperature	K	Process thermal condition.
Rotational Speed	rpm	Machine operating speed.
Torque	Nm	Mechanical load.
Tool Wear	min	Accumulated tool wear.

Additional contextual parameter: Product Type

The dataset contains Type values L, M and H. Type is not a sensor reading, but it is retained as context because the AI4I Overstrain Failure rule has different thresholds for L, M and H.

Target labels

- **Machine Failure:** overall machine-failure target.
- **TWF:** Tool Wear Failure.
- **HDF:** Heat Dissipation Failure.
- **PWF:** Power Failure.
- **OSF:** Overstrain Failure.
- **RNF:** Random Failure.

Important: Failure labels are targets used during training/evaluation. They are not inputs available when the system predicts a new machine condition.

7. Derived Calculations

The five raw parameters are used to calculate additional indicators that help the system interpret machine behavior.

7.1 Temperature Difference

Temperature Difference = Process Temperature - Air Temperature

This measures the thermal gap between the process and the surrounding air and is directly used in the AI4I Heat Dissipation Failure condition.

7.2 Mechanical Power

Power (W) = Torque x RPM x 2 x pi / 60

This converts rotational speed and torque into approximate mechanical power and is used in the AI4I Power Failure condition.

7.3 Overstrain Measure

Overstrain Measure = Tool Wear x Torque

This combines accumulated tool wear and mechanical load and is used in the AI4I Overstrain Failure condition.

7.4 Tool Wear Condition

Tool Wear is already a direct input. In the AI4I generation logic, Tool Wear Failure is associated with a tool-wear failure time between 200 and 240 minutes.

8. Exact AI4I Failure Conditions

These are dataset-specific conditions and should not be replaced with arbitrary project thresholds.

Heat Dissipation Failure (HDF)

Temperature Difference < 8.6 K AND Rotational Speed < 1380 rpm

Power Failure (PWF)

Power < 3500 W OR Power > 9000 W

Overstrain Failure (OSF)

Product Type	OSF condition
L	Tool Wear x Torque > 11,000
M	Tool Wear x Torque > 12,000
H	Tool Wear x Torque > 13,000

Tool Wear Failure (TWF)

Associated with a randomly selected tool-wear failure time between 200 and 240 minutes in the dataset generation logic.

Random Failure (RNF)

Defined as a random failure with a 0.1 percent chance. It has no deterministic sensor threshold, so the system should not force a sensor-based root cause for RNF.

9. Failure Modes / Likely Root Causes

Failure mode	Main evidence	Maintenance focus
Tool Wear Failure	High/critical tool-wear condition.	Inspect or replace the tool.
Heat Dissipation Failure	Temperature difference below 8.6 K and RPM below 1380.	Inspect cooling and heat-dissipation conditions.
Power Failure	Calculated power below 3500 W or above 9000 W.	Inspect load, power delivery and drive system.
Overstrain Failure	Tool Wear x Torque exceeds Type-specific threshold.	Inspect tool and excessive mechanical load.
Random Failure	No deterministic sensor pattern.	Perform broader inspection; low causal certainty.

These five categories directly correspond to the AI4I failure-mode labels. In the application, it is better to say likely failure mode or likely root cause rather than claiming that a sensor rule proves a physical root cause.

10. How We Identify the Likely Cause

Step 1: Read the five machine parameters.

Step 2: Calculate temperature difference, mechanical power and overstrain measure.

Step 3: Apply the relevant AI4I failure conditions.

Step 4: Use the machine-learning model to estimate overall failure probability.

Step 5: Predict or rank the possible failure modes.

Step 6: Explain the prediction using the important parameters and supporting conditions.

Step 7: Recommend the maintenance action associated with the most likely failure mode.

Example

Failure Risk: 82 percent

Likely Failure Mode: Overstrain Failure

Key Evidence: High torque + high tool wear

Recommended Action: Inspect tool condition and machine load.

11. Predictive AI Layer

The condition engine and the machine-learning model have different jobs.

Layer	Purpose
Condition engine	Uses calculations and known rules to detect abnormal conditions.
ML prediction	Learns historical patterns and estimates future failure risk.
Failure-mode prediction	Ranks the likely failure types.
Explainability	Shows which parameters contributed to the prediction.
Recommendation	Converts the predicted failure mode into a maintenance action.

Training

Historical machine parameters + Type -> historical failure labels -> trained model

Prediction

New machine parameters + Type -> model -> failure probability + failure-mode probabilities

Explanation

Prediction -> important parameters -> condition evidence -> human-readable reason

12. Running Time, Idle Time and Downtime

Running time, idle time and downtime are operational metrics. They should be treated separately from the predictive failure model.

For a live manufacturing deployment, a timestamped sensor stream can be used to calculate machine states such as running, idle and down. The AI4I 2020 dataset itself should not be presented as a complete timestamped production log from which exact real-world downtime duration is automatically available.

State	Meaning
Running	Machine is actively operating.
Idle	Machine is available but not actively producing.
Down	Machine is stopped because of a confirmed fault, maintenance or failure.

Key distinction: Condition analysis helps explain abnormal behavior; predictive maintenance estimates future failure risk; downtime analytics measures operational time.

13. Complete System Flow

1. Data Input

Air Temperature + Process Temperature + RPM + Torque + Tool Wear + Type

2. Data Preparation

Clean data, handle missing values, encode Type and prepare training/testing data.

3. Condition Engine

Calculate temperature difference, mechanical power and overstrain measure.

4. Failure Condition Analysis

Apply the AI4I HDF, PWF, OSF and TWF rules and account for RNF as random.

5. Predictive Model

Estimate overall machine-failure probability and rank failure modes.

6. Explainability

Show the important parameters and the conditions supporting the prediction.

7. Recommendation Engine

Map the predicted failure mode to a maintenance action.

8. Dashboard / Agent

Present health, risk, failure mode, evidence and recommended action.

Final flow: Machine data -> Calculations -> Conditions -> Prediction -> Failure mode -> Explanation -> Maintenance

14. Dashboard / Agent Output

Section	Information
Machine Health	Normal / Warning / High Risk / Critical.
Sensor Values	Air temperature, process temperature, RPM, torque and tool wear.
Derived Values	Temperature difference, power and overstrain measure.
Failure Risk	Predicted probability of failure.
Failure Mode	TWF, HDF, PWF, OSF or RNF.
Explanation	Important parameters and supporting conditions.
Recommendation	Suggested maintenance action.
History	Past predictions and parameter trends.

15. Maintenance Recommendation Logic

Failure mode	Suggested action
TWF	Inspect or replace the tool.
HDF	Inspect cooling and heat-dissipation conditions.
PWF	Inspect load, power delivery and drive system.
OSF	Inspect tool and excessive mechanical load.
RNF	Perform broader inspection; do not claim a specific sensor cause.

16. Implementation Plan

Phase	Work
1	Load and inspect the AI4I dataset.
2	Select the five core parameters and retain Type as context.
3	Implement the dataset-specific condition calculations.
4	Train a model for overall Machine Failure.
5	Predict/rank TWF, HDF, PWF, OSF and RNF.
6	Add explainability.
7	Build maintenance recommendation mapping.
8	Build the dashboard/agent.
9	Evaluate precision, recall, F1-score, confusion matrix and failure-mode performance.

17. What We Should Not Claim

- The five sensor parameters alone do not physically prove every root cause.
- The AI4I dataset is not a complete real-time downtime log.
- Failure labels must not be used as prediction-time inputs.
- Random Failure should not be given a forced sensor explanation.
- Arbitrary thresholds should not replace dataset-defined failure rules.
- A model prediction is a probability/risk estimate, not absolute certainty.

18. Final Project Definition

Sector: Manufacturing, kept generic.

Problem: Unexpected machine failures interrupt production and increase operational risk.

Users: Factory managers, maintenance teams, production supervisors, machine operators and plant management.

Solution: A Predictive Maintenance Agent that analyzes machine parameters, detects abnormal behavior, predicts failure, identifies the likely failure mode, explains the result and recommends maintenance.

Core inputs: Air Temperature, Process Temperature, Rotational Speed, Torque and Tool Wear.

Context: Product Type L/M/H.

Derived calculations: Temperature Difference, Mechanical Power and Overstrain Measure.

Failure modes: Tool Wear Failure, Heat Dissipation Failure, Power Failure, Overstrain Failure and Random Failure.

Final output: Failure probability + likely failure mode + evidence + maintenance recommendation.

19. Simple Example

Suppose a machine has high torque and high tool wear.

First, the system calculates the Overstrain Measure by multiplying Tool Wear by Torque.

Next, it checks the threshold for the machine's Product Type.

The ML model also examines the complete pattern of temperature, RPM, torque and tool wear learned from historical records.

If the model predicts high failure risk and Overstrain Failure receives the highest failure-mode probability, the agent reports Overstrain Failure as the likely failure mode.

The agent then explains the important indicators and recommends inspecting the tool and machine load.

20. Final One-Line Concept

Use five machine parameters to understand machine condition, calculate meaningful indicators, detect abnormal behavior, learn historical failure patterns, predict future failure, identify the likely failure mode, explain why, and recommend maintenance.

21. Source Basis

Primary dataset: Predictive Maintenance Dataset (AI4I 2020), Stephan Matzka, Kaggle.

Dataset reference: AI4I 2020 Predictive Maintenance Dataset, UCI Machine Learning Repository.

Scope decision: The project is kept at the generic manufacturing-sector level because the problem statement does not require restriction to one manufacturing sub-sector.

Important dataset note: The AI4I data is synthetic and represents a milling-process environment. It is being used as a representative predictive-maintenance dataset for the broader manufacturing problem.